Project Design Phase
Solution Architecture

Date	30 September 2026
Team ID	SWTID-2026-2365
Project Name	AI FitTrack API – Fitness Tracking API with Google Gemini AI
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram – AI FitTrack API:

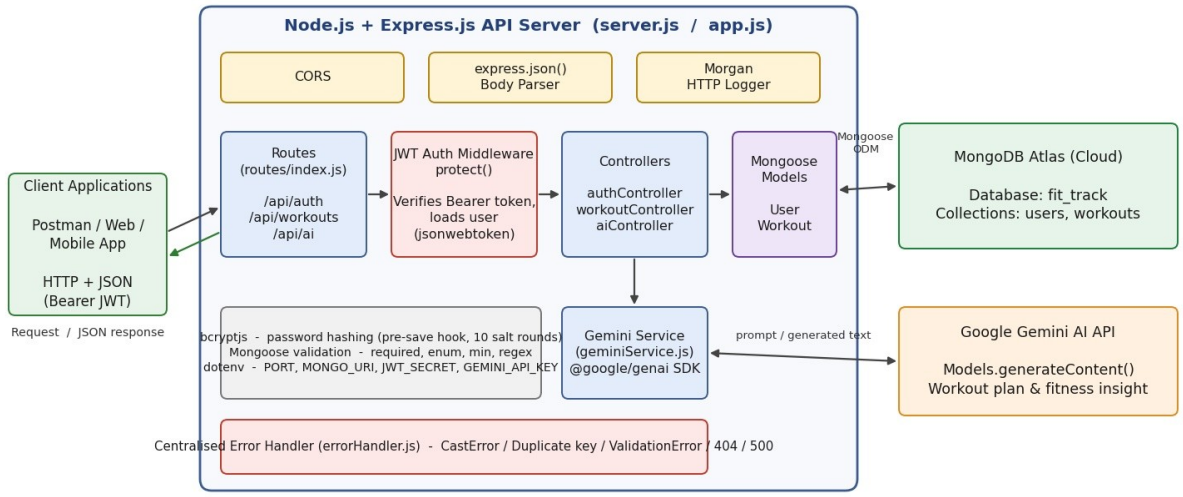


Figure 1: Architecture and data flow of the AI FitTrack API (client → Express server → MongoDB Atlas / Google Gemini AI)

Architecture Description:

Layer / Component	Responsibility
Client Applications	Postman, web or mobile apps send HTTP + JSON requests; private requests carry the header Authorization: Bearer <JWT>.
Express Server (server.js, app.js)	Loads environment variables, connects to MongoDB, enables CORS, JSON body parsing and Morgan request logging, and mounts all routes under /api.
Routes	/api/auth (register, login, profile), /api/workouts (CRUD and search) and /api/ai (workout-recommendation, fitness-insights).
JWT Auth Middleware	protect() verifies the Bearer token with jsonwebtoken, loads the user (without password) and blocks unauthenticated access to workout and AI routes.
Controllers	authController, workoutController and aiController validate input, apply business logic and build JSON responses.
Models (Mongoose)	User (name, unique email, bcrypt-hashed password) and Workout (name, category, duration, calories, date, owner) with validation rules.
Gemini Service	geminiService.js builds prompts and calls Google Gemini through the @google/genai SDK for the recommendation and the insight.
Error Handler	Converts CastError, duplicate-key, validation and other errors into a uniform JSON error response.
MongoDB Atlas / Gemini API	Cloud database for users and workouts; external generative-AI service for personalised text.